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Design Google Drive

In recent years, cloud storage services such as Google Drive, Dropbox, Microsoft OneDrive, and Apple iCloud have become very popular. In this chapter, you are asked to design a cloud storage system.

Let us take a moment to understand Google Drive. Google Drive boasts billions of users. Google Drive is a file storage and synchronization service that allows users to store documents, photos, videos, and other files in the cloud. You can access your files from any computer, smartphone, and tablet. You can easily share files with friends, family, and coworkers [1]. Figure 1 and Figure 2 show user storage size and the number of files and folders per user.

Figure 1

Figure 2

Step 1 - Understand the problem and establish design scope

Designing a Google drive is a big project, and it is important to ask questions to narrow down the scope.

Candidate: What are the most important features?
Interviewer: Upload and download files, file sync, and notifications.
Candidate: Is this a mobile app, a web app, or both?
Interviewer: Both.
Candidate: What are the supported file formats?
Interviewer: Any file type.
Candidate: Do files need to be encrypted?
Interviewer: Yes. Files in the storage must be encrypted.
Candidate: Is there a file size limit?
Interviewer: Yes. Single file size cannot be 10 GB or smaller.
Candidate: How many users do the product have?
Interviewer: 10M DAU.

In this chapter, we focus on the following features:

- Add files. The easiest way to add a file to a drive is by dragging a file into Google drive.
- Download files.
- Sync files across multiple devices. When a file is added to one device, it is automatically synced to other devices.
- See file revisions.
- Share files with your friends, family, and coworkers.
- Send notifications when a file is added, edited, or shared with you.

Features not discussed in this chapter include:

- Google docs editing and collaboration. Google doc allows multiple people to edit the same document simultaneously. This is a part of our design interview.

Non-functional requirements are also important. We discuss the following:

- Reliability. Reliability is important for file storage. We should not lose any user's files.
- Fast sync speed. If a file is edited, the other edits should be propagated and reflected on other devices.
- Bandwidth usage. If a product takes a lot of unnecessary network bandwidth, users will be unhappy, especially when they are on a mobile data plan.
- Scalability. The system should be able to handle high volumes of traffic.
- High availability. Users should still be able to use the cloud storage, even some services are offline, slowed down, or have unexpected network errors.

Back of the envelope estimation

- Assume the application has 50 million users and 10 million DAU.
- Users get 10 GB free space.
- Assume users upload 2 files per day. The average file size is 500 KB.
- 1:1 read to write ratio.
- Cloud storage cost is around 0.02$ / GB / MONTH
- QPS for upload API: 10 million * 2 uploads / 24 hours / 3600 seconds = ~ 240
- Peak QPS: QPS * 2 = ~ 480

Step 2 - Propose high-level design and get buy-in

Instead of covering the high-level design diagram from the beginning, we will use a slightly different approach. We will start with something simple. Build something on a single server. Then, gradually add more components to fix or support shortcomings of users. By doing this exercise, it will refresh your memory about some important topics covered in the course.

Let us start with a single server setup as listed below:

- A web server to upload and download files.
- Relational database management system (RDBMS). We use MySQL, Postgre SQL, Oracle, etc.
- A storage system to save files. We allocate 1TB of storage space on the hard disk to store files.

We spend a few hours setting up an application server, a database, and a directory called drive on the root directory to store uploaded files. Under the directory, there is a list of directories, one for each user. Each user has a directory to store all his/her files. For that user, The filename on the server is kept the same as the original file name. Each file in folder can be uniquely identified by joining the hierarchical path to the file with a delimiter.

Figure 3 shows an example of how the directories look like. The directory structure is organized in a hierarchical way.

Figure 3

APIs

What do the APIs look like? We primarily need 3 APIs: upload a file, download a file, and get file revisions.

1. Upload a file to Google Drive

Two types of uploads are supported:

- Simple upload. Use this upload type when the file size is small.
- Resumable upload. Use this upload type when a file size is large and there's a high chance of network interruption.

Post to an example of resumable upload API:

https://api.example.com/files/upload?uploadType=resumable&

Params:

- uploadType=resumable
- data: could be file to be uploaded.

A resumable upload is achieved by the following 4 steps [2]:

1. Send the initial request to retrieve the resumable URI.
2. Upload the data and monitor upload state.
3. If upload is disrupted, resume the upload.
4. Finish the upload.

2. Download a file from Google Drive

Example API: https://api.example.com/files/download

Params:

- path: download file path.

Example params:

```
path: "/recipes/soup/beef_stew.txt"
```

3. Get file revisions

Example API: https://api.example.com/files/list_revisions

Params:

- path: The path to the file you want to get the revision history.
- limit: The maximum number of revisions to return.

Example params:

```
path: "/recipes/soup/beef_stew.txt",
limit: 10
```

All the APIs require user authentication and use HTTPS. Secure Socket Layers (SSL) protects data transfer between the client and backend servers.

Move away from single server

As more files are uploaded, eventually you get the space full alert as shown in Figure 4.

Figure 4

Only 15 MB of storage space is left. We plan to upgrade our system storage space. We plan to upgrade our system storage space. The first solution comes to mind is to shard the data, so it is stored on multiple storage servers. Figure 5 shows an example of sharding based on user_id.

Figure 5

You pull an alright-sized car and go database sharding and shards it evenly. Everything works smoothly again. You have dropped the file, and you will upload your personal drive backup in case of a disaster. Your drive is full again. You pick up your backpack and head to the trail you that many hiking companies like Netflix and Airbnb use Amazon S3 for storage. Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance[3]. You decide to do the same based on your S3 experience.

After a lot of reading, you get a hang of understanding the S3 storage system and decide to store files in S3. Amazon S3 supports some region and storage replication. A region is a geographical area where Amazon data centers are located. Data does centers. As shown in Figure 6, data can be replicated on the same region (left side) and cross-region (right side). Redundant files are stored in multiple regions to guard against data loss and ensure availability. A bucket is like a folder in the system.

Figure 6

Same-region replication

Cross-region replication

Figure 6

After putting files in S3, you can finally have a good night's sleep without worrying about data loss. To stop similar problems from happening in the future, you decide to do further research on what you can improve. Here are a few ways you find:

- Load balancer. Add a load balancer to distribute network traffic. A load balancer ensures evenly distributed traffic, and if a web server goes down, it redirects traffic to other servers.
- Web servers. After a load balancer is added, more web servers can be added/removed easily depending on the traffic load.
- Metadata database. Move the database out of the server to avoid single point of failure. In the meantime, set up data replication and sharding to meet the available and scalability requirements.
- File storage. Amazon S3 is used for file storage. To ensure availability and durability, files are replicated in two separate geographical regions.

After applying the above improvements, we have successfully decoupled web servers, metadata database, and file storage from a single server. The updated design is shown in Figure 7.

Figure 7

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Sync conflicts

For a large storage system like Google Drive, sync conflicts happen from time to time. When two users modify the same file or folder at the same time, sync conflicts happen. How should we resolve sync conflicts? Google drive adopts a simple and powerful approach and the server that processes the first request to conflict. Figure 8 shows an example of a sync conflict.

Figure 8

While multiple users are editing the same document at the same, it is challenging to keep the document synchronized. Interested readers should refer to the reference material [4] [5].

High-level design

Figure 10 illustrates the proposed high-level design. Let us examine each component of the system.

Figure 10

User: A user uses the application either through a browser or mobile app.

Block servers: Block servers upload blocks to cloud storage. Block storage, referred to as block level storage, is a technology to store data files as separate blocks. Each block has a unique address. It can be quick and efficient. Block servers, with a unique block ID assigned to each block, upload the block to cloud storage. Block servers compress, encrypt, and split files and chunks to cloud storage system [6]. To reconstruct a file, blocks are joined in the order specified. Block servers may also need to implement complicated algorithms to handle sync conflicts, which will be discussed later in this chapter.

Cloud storage: A file is split into smaller blocks and stored in cloud storage.

Cold storage: Cold storage is a computer system designed for storing inactive data, meaning files are not accessed for a long time.

Load balancer: A load balancer evenly distributes requests across API servers.

API servers: These are responsible for almost everything other than uploading files. API servers are used for user authentication, managing user profile, updating file metadata, etc.

Metadata database: It stores metadata of users, files, blocks, versions, etc. Please note that files are stored in the cloud and the metadata database only contains metadata.

Metadata cache: Some of the metadata are cached for fast retrieval.

Notification servers: It is a publisher-subscriber system that allows a client to be transferred from notification service to clients on certain events. For example, when a file is added, edited, or deleted elsewhere, notification servers inform a client. Long polling and websocket are used for notifications. We choose to use long polling because it is simple to implement.

Offline backup queue: If a client is offline or cannot pull the latest changes, the offline backup queue stores the info so changes will be synced when the client is online.

We have discussed the design of Google drive at a high level. Some of the components are complicated and worth careful examination and discussion. Let's dive into some of the deep dive.

Step 3 - Design deep dive

In this section, we will take a look at the following: block servers, metadata database, upload flow, download flow, notification service, cold storage space and block heuristic.

Block servers

Our large files are upload quickly, sending the whole file on each update consumes a lot of bandwidth. Two optimizations are proposed to minimize the amount of network traffic and latency:

- Delta sync. When a file is modified, only modified blocks are transferred instead of the whole file using a sync algorithm [7] [8].
- Compression. Applying compressed on blocks can significantly reduce the data size. Thus, blocks are compressed using compression algorithms depending on file types. For example, gzip and bzip2 are used for compressing plain text files. Different sync algorithms are discussed in stages and videos.

In our system, block servers are the heavy work horse for speeding files. Block servers process files passed from clients by adding a file hash, splitting into blocks, compressing them, and encrypting them. Instead of uploading the whole file to the storage system, only modified blocks are transferred.

Figure 11 shows how a block server works when a new file is added.

Figure 11

- A file is split into smaller blocks.
- Each block is compressed using compression algorithm.
- To ensure security, each block is encrypted before it is sent to cloud storage.
- Blocks are uploaded to the cloud storage.

Figure 12 illustrates delta sync, meaning only modified blocks are transferred to cloud storage. Highlighted blocks "block 1" and "block 3" represent changed blocks. Using delta sync, only these two blocks are uploaded to the cloud storage.

Figure 12

Block servers also a lot more network traffic by uploading delta sync and compression.

High consistency requirement

Our system requires strong consistency by default. It is unacceptable for a file to be shown differently for different clients at the same time. To achieve strong consistency, we must synchronize changes for metadata cache and database layers.

Memcache, a popular open source caching system, by default, which means different replicas might have different data. To achieve strong consistency, we must ensure the following:

- Stale in cache replicas will be invalid.
- Invalidate cache on database write to ensure cache and database hold the same value.

Achieving strong consistency in a distributed system is very expensive. It slows down high scalability. Consistency, Isolation, Durability properties, DBMS transactions, NoSQL databases do not support ACID properties by default. ACID properties must be asynchronously implemented in distributed NoSQL or cloud storage. Join transaction implemented in NoSQL or cloud storage is expensive.

Metadata database

Figure 13 shows the database schema design. Please note this is a highly simplified version as it only includes the most important tables and interesting fields.

Figure 13

User: The user table contains basic information about the user such as username, email, profile photo, etc.

Device: Keeps track device info. Push id is used for sending push notifications to a specific mobile and web application. Please raise a user if you have questions about push notifications.

Namespace: A namespace is the root directory of a user.

File: File table stores everything related to the latest file.

File_version: It stores version history of the. Existing rows are read-only to keep the integrity of the file version history.

Block: A block uniquely identifies a chunk of data. A file of any version can be reconstructed by joining all the blocks in the correct order.

Upload flow

Let us discuss what happens when a client uploads a file. To better understand the flow, we draw the sequence diagram as shown in Figure 14.

Figure 14

In Figure 14, two requests are sent in parallel: add file metadata and upload files to cloud storage. Both requests originate from client 1.

Add file metadata

1. Client 1 sends a request to add the metadata of the new file.
2. Store the new file metadata in metadata DB and change the file status value to "pending."
3. Notify the notification service that a new file is being added.
4. The notification service notifies relevant clients (client 2) that a file is being uploaded.

File content upload

1. Client 1 uploads the content of the file to block servers.
2. Block servers chunk the files into blocks, compress, encrypt the blocks, and upload them to cloud storage.
3. Once the file is uploaded, cloud storage triggers upload completion callback. The request is sent to API servers.
4. The file status changes to "uploaded" in Metadata DB.
5. Notify the notification service that a file is changed to "uploaded."
6. The notification service notifies relevant clients (client 2) that a file is fully uploaded.

Where a file is edited, the flow is similar, so we will not repeat it.

Download flow

Download flow is triggered when a file is added or edited elsewhere. How does a client know if a file is added or edited by another client? There are two ways a client can know:

- If a file is added or edited, be changed by another client, notification service will inform client 1 that changes are made somewhere on it needs to pull the latest data.
- If a client 1 is offline while a file is changed by another client, data will be saved to the cache. When the offline client is online again, it pulls the latest changes.

Once a client knows a file is changed, it first requests metadata via API servers, then downloads blocks to construct the file. Figure 15 shows the diagram for the flow when a client is informed that changes are made to a file. The request sent to server is "pull changes."

Figure 15

1. Notification service informs client 2 that a file is changed somewhere else.

2. Once client 2 knows that new updates are available, it sends a request to fetch metadata.

3. API servers call metadata DB to fetch metadata of the changes.

4. Metadata is returned to the API servers.

5. Client 2 gets the metadata.

6. Once the client receives the metadata, it sends requests to block servers to download blocks.

7. Block servers first download blocks from cloud storage.

8. Cloud storage returns blocks to the block servers.

9. Client 2 downloads all the new blocks to reconstruct the file.

Notification service

To maintain file consistency, any mutation of a file performed locally needs to be informed to other clients to reduce conflicts. Notification service is built to serve this purpose. At the high-level, notification service allows data to be transferred to clients as events happen. Here are a few options:

- Long polling. Dropbox uses long polling [10].
- WebSocket. WebSocket provides a persistent connection between the client and the server. Communication is bi-directional.

Even though both options work well, we opt for long polling for the following two reasons:

- Communication for notification service is not bi-directional. The server sends information about file changes to the client, but not vice versa.
- WebSocket is suited for real-time bi-directional communication such as a chat app. For Google Drive, notifications are sent infrequently with no burst of data.

With long polling, each client establishes a long poll connection to the notification service. If changes to a file are detected, the client will close the long poll connection. Closing the connection means a client must connect to the metadata server to download the latest changes. After a response is received or connection timeout is reached, a client immediately sends a new request to keep the connection open.

Save storage space

To support file version history and ensure reliability, multiple versions of the same file are stored across multiple data centers. Storage space can be filled up quickly with frequent backups of all file revisions. Three techniques are proposed to reduce storage costs:

- De-duplicate data blocks. Eliminating redundant blocks at the account level is an easy way to save space. Two blocks are identical if they have the same hash value.
- Adopt an intelligent data backup strategy. Two optimization strategies can be applied:
- Set a limit: We can set a limit for the number of versions to store. If the limit is reached, the oldest version will be replaced with the new version.
- Keep valuable versions only: Some files might be edited frequently. For example, saving every edited version for a heavily modified document could mean the file is saved over 1000 times within a short period. To avoid unnecessary copies, we could limit the number of saved versions. We give more weight to recent versions. Experimentation is helpful to figure out the optimal number of versions to save.
- Moving infrequently used data to cold storage. Cold data is the data that has not been active for months or years. Cold storage like Amazon S3 glacier [11] is much cheaper than S3.

Failure Handling

Failures can occur in a large-scale system and we must adopt design strategies to address these failures. Your interviewer might be interested in hearing about how you handle the following system failures:

- Load balancer failure: If a load balancer fails, the secondary would become active and pick up the traffic. Load balancers usually monitor each other using a heartbeat, a periodic signal sent between load balancers. A load balancer is considered as failed if it has not sent a heartbeat for some time.
- Block server failure: If a block server fails, other servers pick up unfinished or pending jobs.
- Cloud storage failure: S3 buckets are replicated multiple times in different regions. If files are not available in one region, they can be fetched from different regions.
- API server failure: It is a stateless service. If an API server fails, the traffic is redirected to other API servers by a load balancer.
- Metadata cache failure: Metadata cache servers are replicated multiple times. If one node goes down, you can still access other nodes to fetch data. We will bring up a new cache server to replace the failed one.
- Metadata DB failure.
- Master down: If the master is down, promote one of the slaves to act as a new master and bring up a new slave node.
- Slave down: If a slave is down, you can use another slave for read operations and bring another database server to replace the failed one.
- Notification service failure: Every online user keeps a long poll connection with the notification server. Thus, each notification server is connected with many users. According to the Dropbox talk in 2012 [6], over 1 million connections are open per machine. If a server goes down, all the long poll connections are lost so clients must reconnect to a different server. Even though one server can keep many open connections, it cannot reconnect all the lost connections at once. Reconnecting with all the lost clients is a relatively slow process.
- Offline backup queue failure: Queues are replicated multiple times. If one queue fails, consumers of the queue may need to re-subscribe to the backup queue.

Step 4 - Wrap up

In this chapter, we proposed a system design to support Google Drive. The combination of strong consistency, low network bandwidth and fast sync make the design interesting. Our design contains two flows: manage file metadata and file sync. Notification service is another important component of the system. It uses long polling to keep clients up to date with file changes.

Like any system design interview questions, there is no perfect solution. Every company has its unique constraints and you must design a system to fit those constraints. Knowing the tradeoffs of your design and technology choices are important. If there are a few minutes left, you can talk about different design choices.

For example, we can upload files directly to cloud storage from the client instead of going through block servers. The advantage of this approach is that it makes file upload faster because a file only needs to be transferred once to the cloud storage. In our design, a file is transferred to block servers first, and then to the cloud storage. However, the new approach has a few drawbacks:

- First, the same chunking, compression, and encryption logic must be implemented on different platforms (iOS, Android, Web). It is error-prone and requires a lot of engineering effort. In our design, all those logics are implemented in a centralized place: block servers.
- Second, as a client can easily be hacked or manipulated, implementing encrypting logic on the client side is not ideal.

Another interesting evolution of the system is moving online/offline logic to a separate service. Let us call it presence service. By moving presence service out of notification servers, online/offline functionality can easily be integrated by other services.

Congratulations on getting this far! Now give yourself a pat on the back. Good job!

Reference materials

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